THE LAWS OF SOFTWARE TEAMS

# The Ringelmann effect

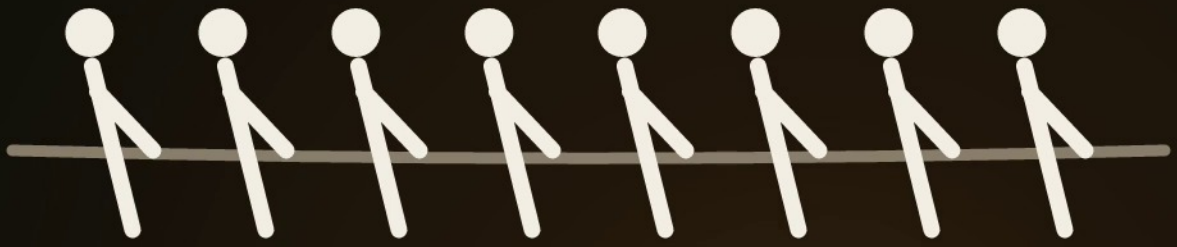
## Individual effort drops as a group gets bigger

swipe →

Ivan Rykov

01 THE HOOK

# 8 people pulled a rope
# They pulled like 4



total pull, in people

of 8

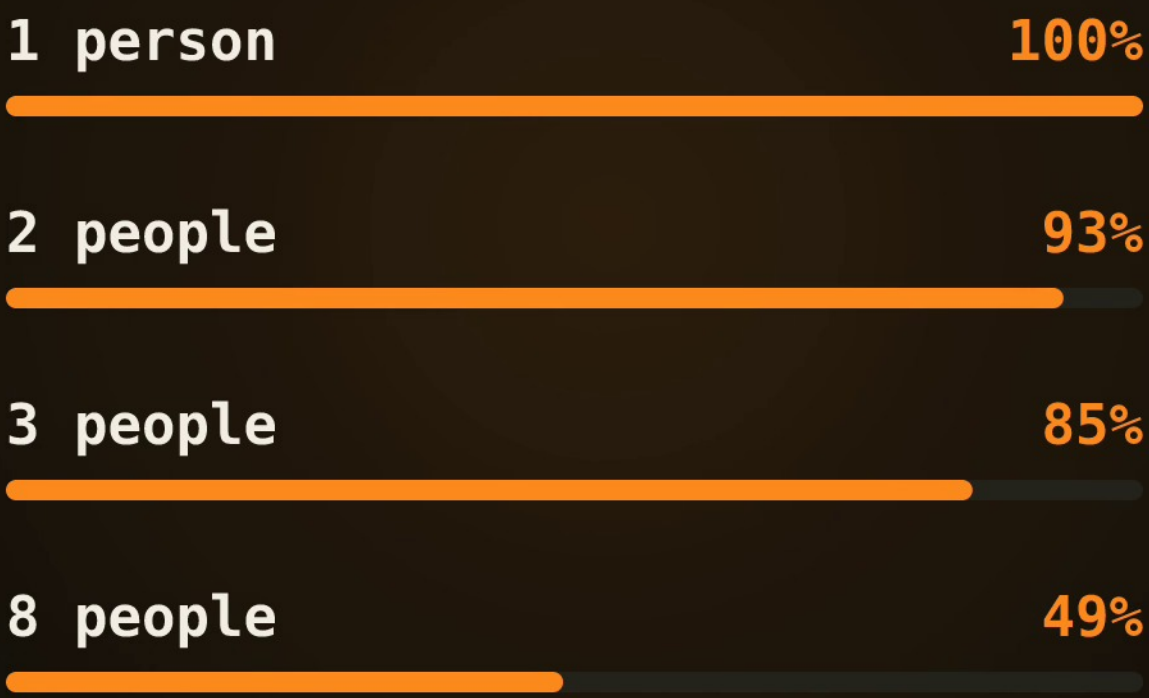
3.9

8 × 49% ≈ 3.9 · Ringelmann, 1913

# Max Ringelmann

French agricultural engineer
Tested in the 1880s, published 1913

effort per person, by group size



# Two things go wrong

1

## Coordination

Nobody pulls at the same moment



2

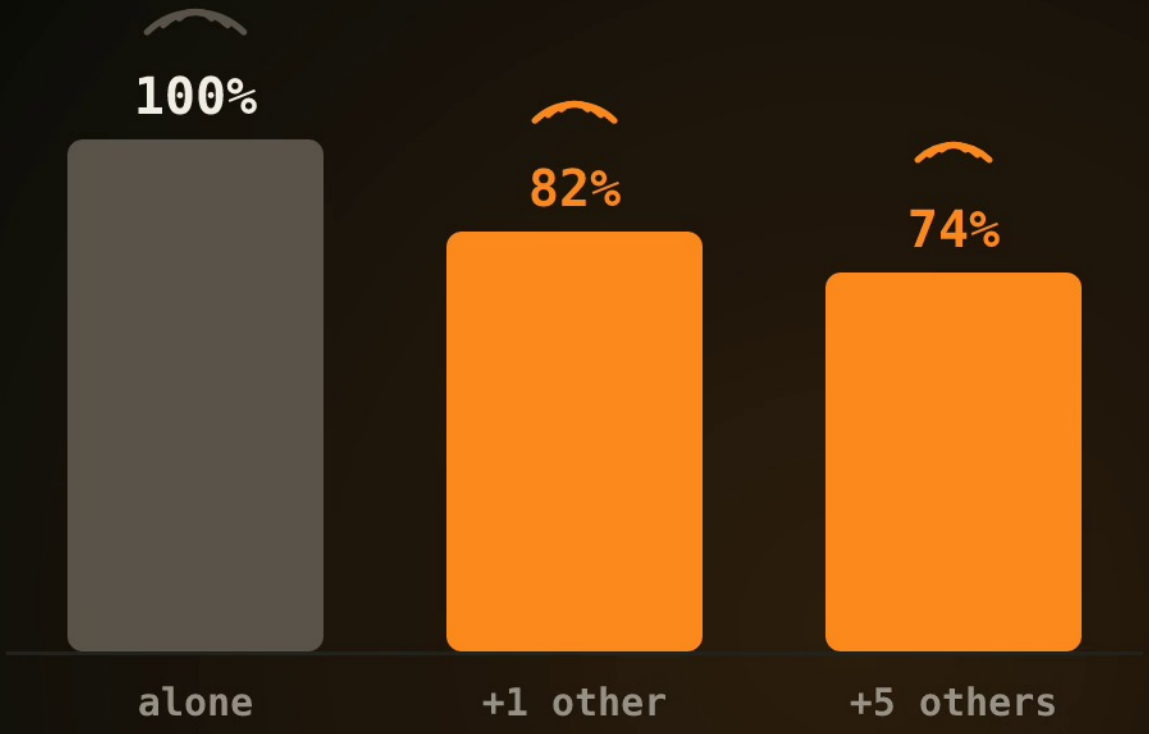
## Motivation

“Someone else will pull”



# 1979: people shouted alone, blindfolded

But told others were shouting too:

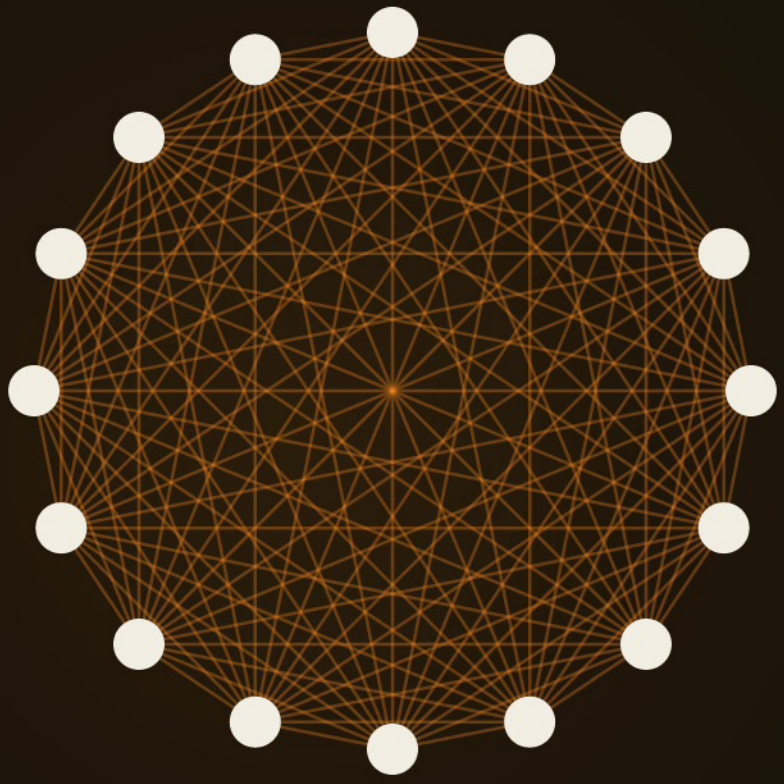



## Nobody to coordinate with
## They just tried less

Latané, Williams & Harkins, 1979 · "social loafing"

05 IN SOFTWARE

# In software, everyone talks to everyone



**16 people → 120 lines**

n(n−1)/2 · Brooks, The Mythical Man-Month, 1975

Ivan Rykov

# The data agrees

390 software projects, same size:
teams of 9+ vs teams under 4

**−30%** schedule

**+350%** cost

**+500%** defects

## Best results: teams of 3-7

491 projects, Putnam 1997

Doug Putnam, QSM

07 THE FIX

# How to beat it:

1. **Keep teams small**
   3-7 people, clear ownership

2. **Make each part visible**
   loafing hides in the crowd

3. **Give work that matters**
   effort follows meaning

Karau & Williams, 1993 · review of 78 studies

**PostHog: 47 people, 15 teams**

via manager.dev

# So what's the right size? Small enough that nobody can hide



TAKEAWAY

# So what's the right size?

# Small enough that nobody can hide

Sources: Kravitz and Martin, 1986 · Latané, Williams and Harkins, 1979 · QSM, Putnam, 1997

next: Price's law